

CHEMICAL KINETICS: MASTER TEXTBOOK

1. Rate of Reaction

Basics

Chemical Kinetics is the branch of chemistry dealing with the rates of reactions and the mechanisms by which they occur.

Rate Law: $\text{Rate} = k[A]^x[B]^y$. The rate of reaction depends on the concentration of reactants.

2. Order and Molecularity

Key Differences

Order: Sum of powers of concentration terms in rate law (determined experimentally).

Molecularity: Number of reacting species taking part in an elementary step (theoretical concept).

3. Arrhenius Equation

Temperature Dependence

The rate of reaction increases significantly with temperature due to an increase in collisions with energy greater than the activation energy.

$$k = A * e^{(-E_a / RT)}$$

Where k is the rate constant, A is the frequency factor, E_a is activation energy, and R is the gas constant.

4. Catalysis

Role of Catalyst

A catalyst increases the rate of reaction by providing an alternative pathway with a **lower activation energy**(E_a).